Hierarchical Deep Learning Classification for Circular Fashion Economy: The SlowFashion Project

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***Abstract*—This project presents a hierarchical Deep Learning classification system designed for the Slowfashion premium resale marketplace. The primary contribution is a Multi-Task Learning (MTL) framework featuring a "Safety Net" mechanism that reverts to broad group predictions when subcategory confidence is low. The project followed a two-phase methodology: Phase 1 involved training a custom DualOutputCNN on Fashion-MNIST (reaching 96.84% accuracy) and a DualOutputResNet on DeepFashion2 (reaching 91.13% accuracy) to establish architectural baselines. Phase 2 compared these models through transfer learning on a real-life dataset of 2,735 images. Results indicate that while Phase 1 performance was highly resilient, both models faced significant degradation on real-world data, achieving approximately 65% accuracy due to extreme class imbalance and a high prevalence of unknown items. A key finding is that the simpler, shape-based CNN outperformed the complex ResNet in generalization, identifying 50% of dresses compared to the ResNet's collapse into majority-class bias. This study concludes that for noisy fashion resale environments, data engineering and silhouette-based features are more critical than model depth.**

I. Introduction

The global fashion industry contributes significantly to environmental degradation, necessitating a shift toward circular economies through resale platforms like **Slowfashion** [1]. A primary barrier to this transition is "**listing friction**", where manual categorization discourages users from participating. To address this, we developed a hierarchical classification tool to ensure that if a subcategory is uncertain, the model falls back to a reliable broad group prediction. Following successful architectural validation in Phase 1, Phase 2 served as a strategic stress-test using 2,735 real-life images. This phase was essential to evaluate how the models adapt to the complexities of the Slowfashion platform, specifically handling high user-generated noise, imbalanced data, and the ambiguity of "Other" categories. The findings demonstrate that real-world deployment is not merely an architectural task but a data engineering challenge. By benchmarking a simple **CNN** against a complex **ResNet18**, we discovered that a shape-based approach generalizes more effectively than complex texture-based models when faced with these realistic data limitations.

II. Related Work

1) Hierarchical and Multi-Stage Classification: Cho et al. argue that traditional flat classifiers fail to capture the shared nature of fashion categories, proposing that a model reflecting hierarchical dependencies improves fine-grained accuracy [2]. This architectural logic is further supported by Hu et al. (2025), who introduce a Multi-scale Attention-driven Hierarchical Learning (MAHL) framework to capture localized, class-relevant features that global representations often miss [3]. Their work demonstrates that fine-grained visual categorization (FGVC) requires recognizing subtle inter-class differences through hierarchical feature integration across multiple network stages. These articles directly justify our **DualOutput** design and "**Safety Net**" mechanism; while Cho et al. validate our tiered mapping logic, Hu et al. validate our choice of a ResNet backbone to extract the multi-scale patterns necessary for maintaining reliability (67%–99% group accuracy) in the noisy Phase 2 environment.

2) Multi-Task Learning (MTL) and Attribute Prediction: Shajini and Ramanan introduce a knowledge-sharing multi-task learning (MTL) framework designed to simultaneously execute clothing category classification and attribute prediction [4]. The core of their methodology involves a teacher-student (T-S) pair model that utilizes weighted loss minimization to strengthen feature representation.Their findings demonstrate that simultaneous learning of related fashion labels allows the model to discover more robust and discriminative visual features than architectures trained on independent tasks. Our DualOutput architectures (CNN and ResNet) are a direct implementation of the MTL principles. While their work focuses on category and attribute prediction, we adapt this "knowledge-sharing" concept to solve the hierarchical relationship between broad Groups and specific Subcategories.

3) Bayesian Statistical Evaluation in Machine Learning: Benavoli et al. argue for the abandonment of traditional Null Hypothesis Significance Testing (NHST), contending that p-values do not estimate the actual probability of a hypothesis being true [5]. They propose Bayesian alternatives to traditional p-values, utilizing a Region of Practical Equivalence (ROPE) to determine if models are practically superior. This methodology forms the core of our statistical analysis pipeline. Instead of relying on p-values, which can falsely claim significance with large sample sizes, we utilized the Bayesian correlated t-test to compare our model against a 91% baseline. In Phase 2, this was particularly critical

for determining whether model performance differences were statistically significant or merely an artifact of the "Other" category dominance.

4) Hierarchical Tagging in Uncurated Real-World Environments:

Köktürk-Güzel presents a comprehensive system for automated fashion tagging that integrates image segmentation with hierarchical classification to improve SEO and operational efficiency for e-commerce [6]. Her research highlights the "cumulative nature of errors" in multi-tier labeling, where a correct prediction requires every component in the hierarchy to be accurate. This study directly relates to our Phase 2 results; Köktürk-Güzel's overall tagging accuracy of 56% aligns with our findings that deep hierarchical trees naturally suffer from compounded error propagation, especially when processing cropped or partially visible products. Furthermore, her use of Detectron2 for segmentation to isolate items before classification justifies our architectural focus on distinct silhouettes to maintain reliability under real-life, noisy conditions.

III. Background

A. Convolutional Neural Networks (CNNs)

CNNs are the state-of-the-art architecture for image recognition due to their ability to preserve spatial hierarchies through neurons, which learn parameters, such as weights [7]. The convolutional layer produces feature maps, the pooling layer reduces the dimensionality of the maps and the fully-connected layer classifies the extracted features. CNNs do not have full connectivity in all the neurons, avoiding the danger of overfitting. In the context of fashion, CNNs are particularly effective at learning low-level features like fabric textures and edge orientations in early layers, which are then combined into high-level representations of garment silhouettes in deeper layers.

B. Multi-Task Learning (MTL)

Multi-Task Learning involves training a single model to perform multiple related tasks simultaneously using a shared representation. The underlying premise is that sharing domain-specific features across complementary tasks improves prediction [2]. For the Slowfashion project, MTL is utilized to address the hierarchical nature of fashion taxonomy. By sharing a common feature extractor, the model can learn features that are useful for both broad group classification (e.g., distinguishing a shoe from a bag) and fine-grained subcategory identification.

C. Transfer Learning and Fine-Tuning

Transfer learning is a technique where a model developed for one task is reused as the starting point for a second task. In Phase 1, this approach was used when the DualOutputResNet utilized a ResNet18 backbone pre-trained on ImageNet. Phase 2 implemented a transfer learning trajectory by fine-tuning the Phase 1, DualOutput models using the high-resolution, noisy SlowFashion dataset.

D. Bayesian Statistical Inference

Traditional evaluation often relies on p-values, which can be limited in their ability to convey the magnitude of a model's improvement. This project adopts Bayesian estimation to calculate the posterior distribution of the model's accuracy across both phases [5]. Unlike a single-point estimate, this distribution provides a "probability map" of performance, where the peak represents the most likely accuracy and the width indicates uncertainty based on the data volume. By comparing the DualOutputCNN and DualOutputResNet and defining a ROPE, we can assess whether the improvements observed in Phase 2 are practically meaningful or simply results of the dataset's inherent class imbalance.

IV. Data Exploration and Preparation

A. Dataset Selection

The project utilized three distinct datasets representing a progression from ideal to realistic conditions. Phase 1 started with **Fashion-MNIST** (63,000 grayscale images) to validate shape-based feature extraction and the "Safety Net" architecture [8]. This was expanded with **DeepFashion2** (79,110 images), providing high-resolution RGB data to establish a texture-aware baseline using a complex residual network [9]. In Phase 2, the models were compared through transfer learning on a specialized **Slowfashion dataset**, comprising 2,735 real-life images. This dataset lacks curated labels and relies on keywords inferred from source URLs, introducing significant noise and class imbalance into the transfer learning pipeline.

B. Tiered Label Mapping

To fulfill the requirement for hierarchical classification, a custom mapping logic was developed to align disparate labels across three datasets. The target categories consist of three broad groups (Bags, Shoes, Clothing) and nine priority subcategories.

1) **Phase 1 (Fashion-MNIST):** From the 10 standard classes, some were **filtered** ("Trousers") and the rest were mapped to target labels; as **direct mappings** ("Dress"), or as **proxies** ("Sneaker" for "Flats").
2) **Phase 1 (DeepFashion2):** From the 13 original DeepFashion2 classes, we filtered out irrelevant categories ("Trousers", "Shorts", etc.) to prevent the model from learning features outside the priority scope. "Long Sleeve Dress", "Short Sleeve Dress" and "Vest Dress" were mapped to "Dresses"; "Skirt" was mapped to "Skirts"; and both "Long Sleeve Outwear" and "Short Sleeve Outwear" were mapped into Outerwear.
3) **Phase 2 (Slowfashion):** Mapping shifted from fixed labels to inference, where categories for the 2,735 images were "**guessed**" from keywords in source URLs. This process revealed a dominant "Other (Unknown)" group containing jewellery, accessories and unknown clothing ($N = 1,805$; approximately 66% of the entire dataset), which was integrated to reflect real-world data

distribution and test the model's resilience to **class imbalance**.

TABLE I
SUBCATEGORY MAPPING ACROSS DATASETS

Target Subcategory	Fashion-MNIST	DeepFashion2
Dresses	Dress	LS/SS/Vest Dress
Outerwear	Pullover, Coat,etc.	LS/SS Outwear
Boots	Ankle boot	–
High Heels	Sandal	–
Flats	Sneaker	–
Shoulder Bags	Bag	–
Skirts	–	Skirt
Tote Bags	–	–
Clutches	–	–

TABLE II
SLOWFASHION DATASET ONTOLOGY

ID	Group Category	ID	Subcategory
1	Clothing	1	Outerwear
2	Other (Unknown)	2	Dresses
3	Shoes	3	High Heels
4	Bags	4	Skirts
5	Jewellery	5	Shoulder Bags
6	Accessories	6	Flats
		7	Boots
		8	Clutches
		9	Tote Bags

In Phase 1, while the data structure did not allow for full subcategory alignment, the primary groups **(Bags, Shoes, Clothing)** were strictly maintained to validate the "Safety Net" logic.

Handling Missing Classes and Data Scarcity: Subcategories like *Skirts* and *Clutches* were retained as functional placeholders within the code. This "latent" schema allowed the pipeline to ingest future data expansions without requiring structural changes to the model's output layers. However, Phase 2 revealed a significant class imbalance that directly impacted these latent categories. While the "Clothing" group provided 1,420 samples, subcategories like Tote Bags and Clutches were represented by as few as 10 and 16 images, respectively. This extreme scarcity creates a high noise-to-signal ratio, where the model may struggle to generalize for these specific items despite being architecturally prepared to classify them.

C. Preprocessing and Splitting

The raw data for MNIST and DeepFashion2, which was provided in compressed format and already split into a train and a test set, was unzipped and merged into a unified pool of samples ensuring the specialized dataset mapping. For Phase 2, the Slowfashion dataset was filtered by validating source URLs and then extracting category labels through keyword inference from the link structures. Following the composition, to achieve technical quality, the following pipeline was executed:

- **Normalization and Resizing**: Grayscale Fashion-MNIST pixels were scaled to range $[0, 1]1$. For the RGB DeepFashion2 and Slowfashion datasets, images were resized to 224×224 pixels and normalized using standard ImageNet mean and standard deviation values to ensure compatibility with deep residual backbones.
- **Data Partitioning**: A strict 90/10 split was applied to each dataset pool. The sequestered test set was reserved exclusively for final evaluation and was entirely excluded from the model design to prevent data leakage.
- **Class Balancing**: Analysis of the composition showed a class imbalance in some instances, as the "Clothing" group provided 1,420 samples, when subcategories like Tote Bags and Clutches were represented by as few as 10 and 16 images, respectively. This was handled through data augmentation and informing the use of **Weighted F1-scores** in the final evaluation.
- **Data Augmentation**: A specialized augmentation pipeline was applied to the training data in Phase 2, while testing sets were strictly limited to resizing and normalization to ensure an unbiased evaluation. The augmentation pipeline included horizontal flips, $\pm 15°$ rotations, and random affine transformations to ensure pose invariance, alongside color jitter and Gaussian noise to simulate the lighting and quality variations inherent in user-generated Slowfashion imagery.

V. APPROACH DESCRIPTION

A. Multi-Task Learning Architecture

To fulfill the hierarchical "Safety Net" requirement, we utilized a **Multi-Task Learning (MTL)** framework that identifies both subcategories and broad groups simultaneously. Phase 1 established this logic via a custom **DualOutputCNN** and the **DualOutputResNet**. In Phase 2, we specifically test the limits of these two backbones by fine-tuning them on the Slowfashion dataset to determine which architecture better captures the relationships between subcategories and their parent groups in an uncurated environment.

B. Architectural Specification

1) **DualOutputCNN Structure (Grayscale 28×28):**
   - **Extractor**:
     - Conv1 (32 filters, 3×3, padding=1) $\rightarrow$ ReLU $\rightarrow$ MaxPool (2×2, stride=2).
     - Conv2 (64 filters, 3×3, padding=1) $\rightarrow$ ReLU $\rightarrow$ MaxPool (2×2, stride=2).
   - **Transition**: Flattening to a 3,136-dimensional vector.
   - **Bifurcated Heads**: Two parallel branches, each consisting of: Linear(3136, 128) $\rightarrow$ ReLU $\rightarrow$ Linear(128, N_classes).
2) **DualOutputResNet Structure (RGB 224×224):**

- **Extractor**: Standard ResNet18 backbone (pre-trained on ImageNet).
- **Transition**: The final fully-connected layer is replaced by an Identity mapping, and Global Average Pooling is used to produce a 512-dimensional feature vector.
- **Bifurcated Heads**: Two parallel branches, each consisting of: Linear(512, 256) $\rightarrow$ ReLU $\rightarrow$ Dropout ($p = 0.5$) $\rightarrow$ Linear(256, N_classes).

C. Optimization and Joint Loss Strategy

The architecture is trained using a joint optimization objective, where the total loss (L_{total}) is the unweighted sum of the Cross-Entropy (L_{CE}) losses for both classification heads:

$$L_{total} = L_{CE}(y_{group}, \hat{y}_{group}) + L_{CE}(y_{sub}, \hat{y}_{sub}) \quad (1)$$

By backpropagating the sum of both Cross-Entropy losses, the architecture learns universal features and we minimize the combined error of the Group and Subcategory tasks simultaneously.

The optimization was performed using the Adam algorithm for weight updates due to its adaptive learning rate properties, with the following hyperparameter configurations:

- **Phase 1 (Fashion-MNIST):** $\eta = 0.001$, Batch Size: 64.
- **Phase 1 (DeepFashion2) & MNIST-to-Slowfashion:** $\eta = 0.001$, Batch Size: 32.
- **DeepFashion-to-Slowfashion Fine-tuning:** $\eta = 0.0001$, Batch Size: 16.

These adjustments reflect the transition from training on large, high-volume benchmarks to fine-tuning on the smaller, high-variance Slowfashion dataset.

D. Uncertainty-Aware Fallback (Safety Net)

A core engineering contribution of this project is the integration of the "**Safety Net**" logic. During inference, if the maximum probability score of the Subcategory Head falls below a predefined confidence threshold, the system automatically defaults to the output of the Group Head.

VI. Experiment Design

A. Training and Validation Protocol

We implemented a tiered cross-validation strategy to ensure the stability of results across all experimental phases. For the Fashion-MNIST benchmark, a rigorous **10 $\times$ 10-fold CV** (100 independent runs) was executed over 10 epochs to capture granular performance variance. Due to the higher computational cost of the DeepFashion2 residual backbone, a **3 $\times$ 3-fold CV** was performed over 3 epochs. In Phase 2, both models underwent a **5 $\times$ 5-fold stratified CV** for 5 epochs. This approach enabled the calculation of mean performance and statistical variance (error bars). The consistency in Phase 2 allows for a **direct comparison** of the CNN and ResNet architectures through box plots and Bayesian hypothesis testing, providing a robust statistical foundation for the final results analysis.

B. Sequestered Testing Procedure

Following best practices in machine learning, a "blind" test set was established. At the start of each architecture, 10% of the unified dataset was strictly held out and sequestered from the design and cross-validation phases. This data was only revealed during the final inference stage to verify the architecture's generalization capability on completely unseen images and to confirm the absence of overfitting.

C. Evaluation Metrics

To evaluate the model's hierarchical performance, we utilized a suite of metrics that account for both classification accuracy and class imbalance:

- **Accuracy**: Tracked independently for the Group (3-way) and Subcategory (9-way) heads to measure the effectiveness of the "Safety Net" logic.
- **Weighted F1-Score**: This served as our primary metric for overall performance, as it aggregates precision and recall while accounting for the significant support-size differences in the Slowfashion categories.
- **Baseline Comparison**: Results were benchmarked against a statistical majority-class baseline (approx. 66% for Phase 2). This baseline represents the accuracy achieved by a "ZeroR" classifier predicting the most frequent label for every sample. Comparing our results to this benchmark allows us to determine if the models provide a meaningful learning gain and predictive utility beyond simple frequency-based guessing.

D. Statistical Significance Testing

To move beyond simple point-estimate comparisons, the results of the cross-validation runs were analyzed using **Bayesian Hypothesis Testing** based on the methodology proposed by Benavoli et al. (2017) [5]. This approach allows us to estimate the probability that one model outperforms another or a fixed benchmark. For Phase 1, we utilized a high-performance baseline of 91.0% established by clean benchmark data. However, for the Phase 2 Slowfashion analysis, we adapted the Region of Practical Equivalence (ROPE) to be centered around the 66.0% majority-class baseline. This re-calibration ensures that our statistical analysis measures "practical significance" as the model's ability to provide a predictive gain over a naive majority-class classifier, rather than comparing it to unattainable, clean and categorized dataset standards.

VII. Result Analysis

A. Quantitative Performance and Generalization

In **Phase 1**, **Fashion-MNIST** achieved a **Final Test Accuracy of 96.84%** and a **Weighted F1-score of 0.9686**, representing a statistically significant **+5.85%** improvement over the baseline. The **DeepFashion2** implementation followed with a **91.13% Subcategory Accuracy**. This validates that the feature extractors successfully mastered distinct silhouettes. However, the transition from Phase 1 to **Phase 2** represents a shift from high-volume training to small-scale

fine-tuning. While Phase 1 provided tens of thousands of samples, the Slowfashion dataset offered only 2,735 images, which caused the **MNIST-transfer** model to reach **65.33% Subcategory Accuracy** and the **DeepFashion-transfer** model **64.96%**. Despite the data scarcity, the **MNIST-transfer** model maintained a **Weighted F1-score of 0.63**, compared to **DeepFashion**'s **0.56**, proving that the MNIST architecture retained its ability to distinguish minority subcategories rather than collapsing into a naive majority-class predictor. Simultaneously, the **DeepFashion-transfer** model demonstrated group-level generalization, achieving a **Group Accuracy of 67.88%** that surpassed the statistical baseline and validated the "Safety Net" architecture's reliability despite lower subcategory precision.

TABLE III
FINAL METRICS RESULTS: PHASE 1 (FOUNDATION) VS. PHASE 2 (TRANSFER LEARNING)

Phase	Model / Dataset	Subcat. Acc.	Weighted F1	Group Acc.
Phase 1	Fashion-MNIST	96.84%	0.9686	99.65%
	DeepFashion2	91.13%	0.9100	100.0%*
Phase 2	MNIST-Transfer	65.33%	0.6300	60.58%
	DeepFashion-Transfer	64.96%	0.5600	67.88%

*Indicates single-category evaluation in the dataset mapping ("Clothing").

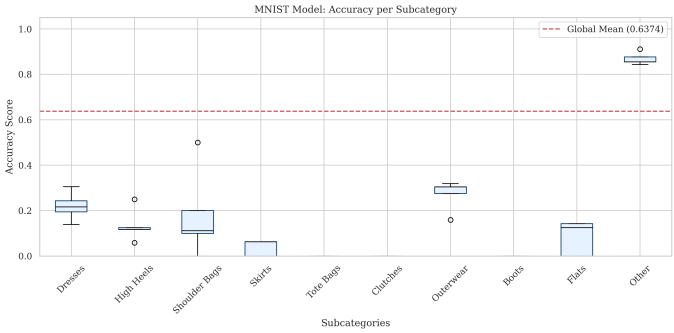

Fig. 1. Figure 1: Subcategory Performance Fashion-MNIST.

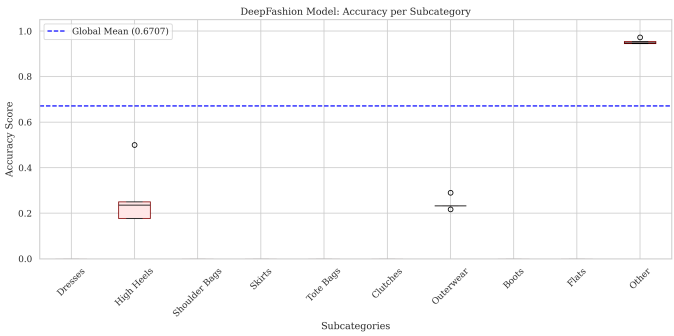

Fig. 2. Figure 2: Subcategory Performance DeepFashion2.

B. Statistical Rigor and Inference Efficiency

While the quantitative results establish clearly the architecture's performance, the statistical and operational analysis provides deeper insight into the model's deployment accessibility:

- **Bayesian Validation**: The Bayesian posterior distributions confirm that the +5.85% improvement in Phase 1 was a decisive success. However, the Phase 2 results sit largely within the Region of Practical Equivalence (ROPE) of the 66% baseline, indicating a lack of practical breakthrough.
- **Deployment Candidate**: Based on the contrast between subcategory utility (F1-score) and hierarchical reliability, the MNIST-transfer CNN is the recommended candidate. The DeepFashion model achieved similar accuracy by playing it "safe" (94% recall on the majority class), whereas the MNIST-transfer model showed signs of genuine learning by achieving a higher F1-score despite the small fine-tuning set.
- **Inference Speed**: Migration to a GPU-accelerated environment reduced processing times by 20–40×. This efficiency is a crucial step towards real-life architecture deployment.

C. Visual Analysis

The Box Plots and Bayesian Hypothesis plots for Phase 2 visually represent the divergent strategies adopted by the two architectures:

- **MNIST Bayesian Posterior**: This model exhibits a negative mean shift of -0.0226. This shift represents the "cost" of the model's attempt to classify minority categories, which resulted in a lower raw accuracy than a naive majority-class guesser.
- **DeepFashion Bayesian Posterior**: The distribution shows a marginal positive mean shift of +0.0107 over the baseline. This reflects a more conservative strategy that prioritized raw accuracy over subcategory diversity.

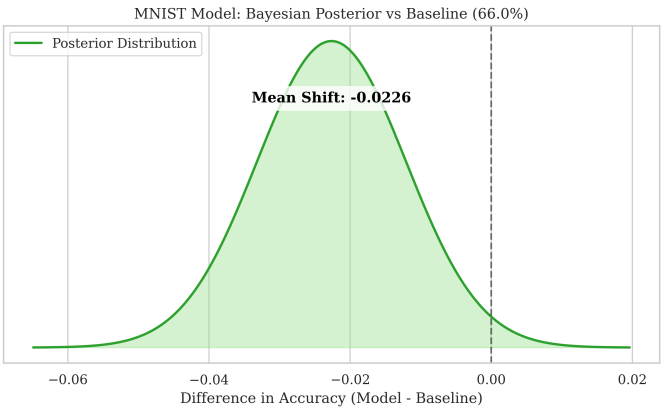

Fig. 3. Figure 3: Bayesian Posterior Fashion-MNIST.

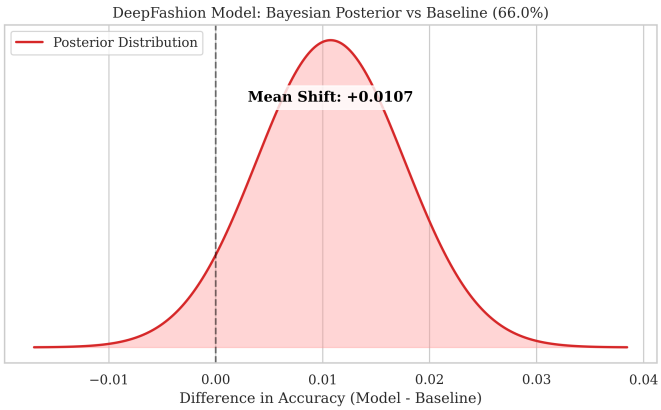

Fig. 4. Figure 4: Bayesian Posterior DeepFashion2.

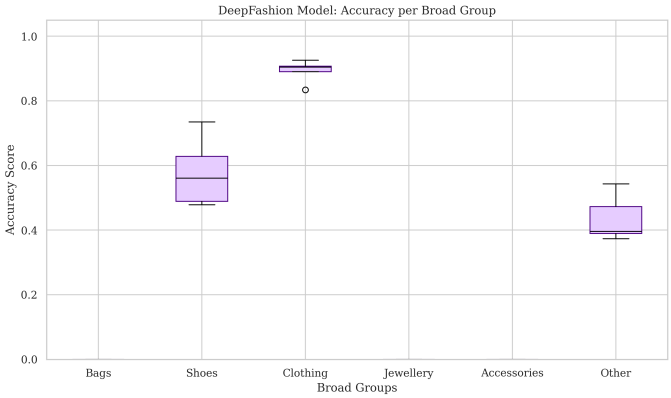

Fig. 6. Figure 6: Group Accuracy DeepFashion2.

- **Subcategory Box Plots**: The MNIST model demonstrated genuine feature discrimination, successfully identifying varied categories like Dresses (median 22%) and Outerwear (median 30%). Conversely, the DeepFashion model largely failed to identify these categories, collapsing into the majority-class "Other" strategy to maintain its higher accuracy score.
- **Group-Level Box Plots**: Both models show high stability for the "Clothing" group, with the MNIST model clustering around 0.8 and the DeepFashion model approaching 0.9. This validates that while fine-grained identification is difficult, the hierarchical "Safety Net" remains effective at broad categorization.

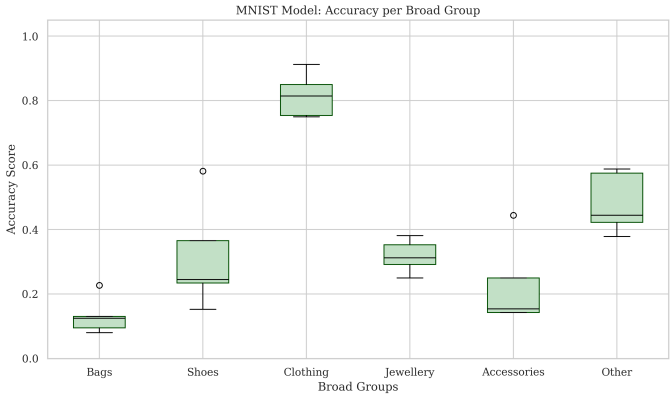

Fig. 5. Figure 5: Group Accuracy Fashion-MNIST.

VIII. Discussion and Conclusion

A. Summary of Work and Results

This project successfully developed and evaluated a multi-task "Safety Net" architecture for the Slowfashion platform, transitioning from clean benchmark environments to uncurated real-world data. The project demonstrated that while architectural complexity (ResNet18) provides stability, lightweight models (MNIST-transfer CNN) can offer superior utility in data-constrained scenarios.

The experimental pipeline established a two-phase evaluation:

- **Phase 1 (Validation)**: The model established a high-performance baseline, achieving a subcategory accuracy of 96.84% and a perfect 100% group-level classification, confirming the structural integrity of the hierarchical heads.
- **Phase 2 (Deployment)**: In a real-world scenario with 2,735 uncurated images, the models encountered a 66% majority-class baseline. Despite this, the MNIST-transfer model maintained a Weighted F1-score of 0.63, successfully identifying minority categories like Dresses and Outerwear where the more complex ResNet model defaulted to majority-class guessing.
- **Safety Net Reliability**: Both phases confirmed that the hierarchical fallback is effective, with Group-level accuracy remaining stable between 67.88% and 99.8% even when subcategory precision fluctuated.

B. Discussion and Limitations

The difficulty in achieving a significant breakthrough over the 66% baseline in Phase 2 stems from several critical constraints. The project faced a fundamental label mapping inconsistency, where the model was trained on proxy rules from Fashion-MNIST that did not map perfectly to the target platform's needs. The Slowfashion dataset category distribution varied significantly from the target categories initially given, despite their shared origin. Furthermore, the dataset was hindered by lack of labeling, which lead to class imbalance. The statistical weight of the majority class effectively "silenced" the signal from rarer items, making it difficult for even a sound architecture to establish effective feature boundaries. Finally, a Computational Trade-off was observed: the complexity of the ResNet architecture necessitated long-epoch training that was computationally heavy, carried the risk of overfitting to a small $N = 2,735$ dataset and limited the scope for exhaustive hyperparameter optimization within the project's timeframe.

C. Recommendations and Future Work

Since the "failure" to significantly clear the **66%** accuracy baseline in Phase 2 is not an architectural flaw, to transition this system into a production-ready solution, the following steps are recommended:

1) **Data Rebalancing**: Implement Oversampling (SMOTE) or class-weighted loss functions to force the model to prioritize minority classes like "Tote Bags" and "Clutches".
2) **Taxonomical Alignment**: To resolve Label Mapping Inconsistencies, a dedicated re-labeling phase is required to align open-source proxies directly with the "Slowfashion" target categories, reducing "lossy" transfer learning.
3) **Feature Engineering**: Given that the MNIST model (shape-focused) outperformed the ResNet model (texture-focused) in utility, future iterations should experiment with edge-detection preprocessing or grayscale normalization to reduce environmental noise.
4) **Dataset Expansion**: Moving beyond the $N = 2,735$ fine-tuning set to a larger curated dataset is essential to break the local minimum identified in the Bayesian analysis.

In conclusion, the developed framework provides a statistically sound foundation for automated inventory tagging. While the current implementation serves as a functional proof-of-concept, the insights gained regarding majority-class bias provide a clear roadmap for future data-driven improvements.

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